CEFET/RJ

PESQUISA EM COMPUTAÇÃO

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Pesquisa primária versus secundária

- Pesquisa primária
  - Trabalho original
  - Dados novos (formulários, entrevistas, observações)
  - Metodologias: qualitativa, quantitativa, analítica

Formalização

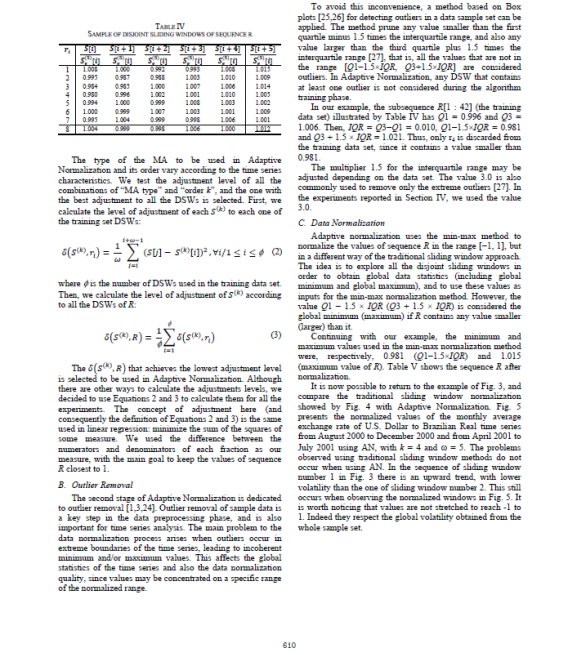

TABLE IV
SAMPLE OF DISJOINT SLIDING WINDOWS OF SEQUENCE R

r_i	$S[i]$ $S^{(k)}[i]$	$S[i+1]$ $S^{(k)}[i]$	$S[i+2]$ $S^{(k)}[i]$	$S[i+3]$ $S^{(k)}[i]$	$S[i+4]$ $S^{(k)}[i]$	$S[i+5]$ $S^{(k)}[i]$
1	1.004	1.000	0.982	0.995	1.008	1.037
2	0.995	0.987	0.988	1.003	1.010	1.009
3	0.984	0.985	1.000	1.007	1.006	1.014
4	0.980	0.996	1.002	1.001	1.010	1.005
5	0.994	1.000	0.999	1.008	1.003	1.002
6	1.000	0.999	1.007	1.003	1.001	1.009
7	0.995	1.004	0.999	0.998	1.006	1.001
8	1.004	0.999	0.988	1.006	1.000	1.012

The type of the MA to be used in Adaptive Normalization and its order vary according to the time series characteristics. We test the adjustment level of all the combinations of "MA type" and "order k", and the one with the best adjustment to all the DSWs is selected. First, we calculate the level of adjustment of each $S^{(k)}$ to each one of the training set DSWs:

$$\delta(S^{(k)}, r_i) = \frac{1}{\omega} \sum_{l=i}^{i+\omega-1} (S[l] - S^{(k)}[i])^2, \forall i / 1 \leq i \leq \phi \quad (2)$$

where ϕ is the number of DSWs used in the training data set. Then, we calculate the level of adjustment of $S^{(k)}$ according to all the DSWs of R:

$$\delta(S^{(k)}, R) = \frac{1}{\phi} \sum_{i=1}^{\phi} \delta(S^{(k)}, r_i) \quad (3)$$

The $\delta(S^{(k)}, R)$ that achieves the lowest adjustment level is selected to be used in Adaptive Normalization. Although there are other ways to calculate the adjustments levels, we decided to use Equations 2 and 3 to calculate them for all the experiments. The concept of adjustment here (and consequently the definition of Equations 2 and 3) is the same used in linear regression: minimize the sum of the squares of some measure. We used the difference between the numerators and denominators of each fraction as our measure, with the main goal to keep the values of sequence R closest to 1.

B. Outlier Removal

The second stage of Adaptive Normalization is dedicated to outlier removal [1,3,24]. Outlier removal of sample data is a key step in the data preprocessing phase, and is also important for time series analysis. The main problem to the data normalization process arises when outliers occur in extreme boundaries of the time series, leading to incoherent minimum and/or maximum values. This affects the global statistics of the time series and also the data normalization quality, since values may be concentrated on a specific range of the normalized range.

To avoid this inconvenience, a method based on Box plots [25,26] for detecting outliers in a data sample set can be applied. The method prune any value smaller than the first quartile minus 1.5 times the interquartile range, and also any value larger than the third quartile plus 1.5 times the interquartile range [27], that is, all the values that are not in the range $[Q1-1.5\times IQR,\ Q3+1.5\times IQR]$ are considered outliers. In Adaptive Normalization, any DSW that contains at least one outlier is not considered during the algorithm training phase.

In our example, the subsequence $R[1:42]$ (the training data set) illustrated by Table IV has $Q1 = 0.996$ and $Q3 = 1.006$. Then, $IQR = Q3-Q1 = 0.010$, $Q1-1.5\times IQR = 0.981$ and $Q3 + 1.5 \times IQR = 1.021$. Thus, only r_{15} is discarded from the training data set, since it contains a value smaller than 0.981.

The multiplier 1.5 for the interquartile range may be adjusted depending on the data set. The value 3.0 is also commonly used to remove only the extreme outliers [27]. In the experiments reported in Section IV, we used the value 3.0.

C. Data Normalization

Adaptive normalization uses the min-max method to normalize the values of sequence R in the range $[-1, 1]$, but in a different way of the traditional sliding window approach. The idea is to explore all the disjoint sliding windows in order to obtain global data statistics (including global minimum and global maximum), and to use these values as inputs for the min-max normalization method. However, the value $Q1 - 1.5 \times IQR$ ($Q3 + 1.5 \times IQR$) is considered the global minimum (maximum) if R contains any value smaller (larger) than it.

Continuing with our example, the minimum and maximum values used in the min-max normalization method were, respectively, 0.981 ($Q1-1.5\times IQR$) and 1.015 (maximum value of R). Table V shows the sequence R after normalization.

It is now possible to return to the example of Fig. 3, and compare the traditional sliding window normalization showed by Fig. 4 with Adaptive Normalization. Fig. 5 presents the normalized values of the monthly average exchange rate of U.S. Dollar to Brazilian Real time series from August 2000 to December 2000 and from April 2001 to July 2001 using AN, with $k = 4$ and $\omega = 5$. The problems observed using traditional sliding window methods do not occur when using AN. In the sequence of sliding window number 1 in Fig. 3 there is an upward trend, with lower volatility than the one of sliding window number 2. This still occurs when observing the normalized windows in Fig. 5. It is worth noticing that values are not stretched to reach -1 to 1. Indeed they respect the global volatility obtained from the whole sample set.

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Avaliação experimental

Adaptive Normalization: A Novel Data Normalization Approach for Non-Stationary Time Series

The length of the sliding windows of NN-SW and the disjoint windows of NN-AN is the same, and equals to the number of inputs plus one (the output), giving to both methods the same dataset size for training. The value of the inertia parameter (order of the used moving average) of NN-AN is at most 10% of the sample size and is chosen by the best level of adjustment of all the possible moving averages, as described in Subsection III-A.

When training the network, the learning rate was set to 0.64, and the back-propagation algorithm also used a momentum of 0.8. The network was trained for 200,000 epochs. This choice was made to give equal conditions to all different normalization techniques and is in agreement with previous work [15] in which this structure usually presented results that were near to the optimal ones.

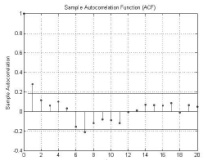

Fig. 6 Sample Autocorrelation of the monthly average exchange rate of U.S. Dollar to Brazilian Real time series

A. Forecasting U.S. Dollar to Brazilian Real Exchange Rate

The first time series used in our experiments was the monthly average exchange rate of U.S. Dollar to Brazilian Real. This time series represents a period of 132 months, varying from January 1999 to December 2009. The training set covered the first 120 months and the test set covered the last 12 months. The number of inputs used in this example was 7. The EMA was used with inertia 8.

Table VI shows the RMSE and MAPE obtained by the six algorithms for the 1-step-ahead and 12-step-ahead forecasts during the test period. The NN-AN achieved the best results for both the 1-step-ahead horizon, with RMSE and MAPE 24% and 14% smaller than the AR (2nd best), and the 12-step-ahead horizon. For the 12-step-ahead horizon, the RMSE and MAPE were 24% and 22% smaller than the NN-SW (2nd best).

B. Forecasting Brazilian Agricultural Gross Product

The second time series used in our experiments was the quarterly Brazilian Agriculture Gross Product - Index Linked (average in 1995 = 100). This time series is seasonally adjusted, i.e., it does not contain the seasonal component. By removing the seasonal component, it is better to reveal certain non-seasonal features and easier to focus on the trend and cyclical components of the time series. The series represents a period of 119 quarters, varying from 1980 Q1 to 2009 Q3. The training set covered the first 107 quarters and the test set covered the last 12 quarters. Both the number of inputs and the SMA order were equal to 4.

TABLE VI
PERFORMANCE OF ALGORITHMS TO FORECAST THE MONTHLY AVERAGE EXCHANGE RATE OF U.S. DOLLAR TO BRAZILIAN REAL TIME SERIES

Algorithm	RMSE 1-step	RMSE 12-step	MAPE (%) 1-step	MAPE (%) 12-step
AR	0.082	0.345	3.174	27.099
NN-MM	0.177	1.173	8.446	57.611
NN-ZS	0.094	1.444	3.545	69.517
NN-ZS	0.126	0.814	4.526	40.931
NN-SW	0.088	0.451	3.661	20.917
NN-AN	0.062	0.345	2.730	18.398

Table VII presents the RMSE and MAPE of the forecasts. As in the previous experiment, the NN-AN achieved the best results considering both RMSE and MAPE and the two forecasting horizons. Considering the 1-step-ahead horizon, the NN-AN obtained both the RMSE (4%) and MAPE (17%) smaller than the AR. For the 12-step-ahead horizon, both the RMSE (3%) and the MAPE (8%) were smaller than the NN-DS.

TABLE VII
PERFORMANCE OF ALGORITHMS TO FORECAST THE QUARTERLY BRAZILIAN AGRICULTURE GROSS PRODUCT TIME SERIES

Algorithm	RMSE 1-step	RMSE 12-step	MAPE (%) 1-step	MAPE (%) 12-step
AR	0.082	0.345	3.174	27.099
NN-MM	0.177	1.173	8.446	57.611
NN-ZS	0.094	1.444	3.545	69.517
NN-ZS	0.126	0.814	4.526	40.931
NN-SW	0.088	0.451	3.661	20.917
NN-AN	0.062	0.345	2.730	18.398

C. Forecasting São Paulo Unemployed Rate

The third experiment used the time series of the monthly Unemployment Rate of São Paulo. This series considers 130 months, varying from January 1999 to October 2009. The first 118 months were chosen for training and the last 12 months were chosen for test. For this experiment, we used 16 inputs and a SMA of order 15.

Table VIII presents the RMSE and MAPE of the forecasts. Again, the NN-AN gave the best results for all the tests, with the RMSE 15% and the MAPE 14% smaller than the one obtained by the AR for the 1-step-ahead forecasts. For the 12-step-ahead horizon, the RMSE (9%) and the MAPE (17%) were smaller than the NN-SW.

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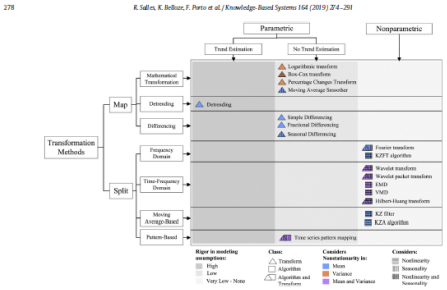
[1] Primary Research vs Secondary Research - https://research.com/research/primary-research-vs-secondary-research

[2] E. Ogasawara, L.C. Martinez, D. De Oliveira, G. Zimbrão, G.L. Pappa, and M. Mattoso, 2010, Adaptive Normalization: A novel data normalization approach for non-stationary time series, In: *Proceedings of the International Joint Conference on Neural Networks*

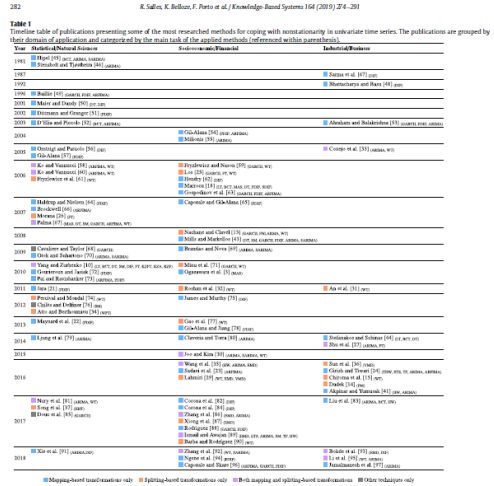
Pesquisa primária versus secundária

- Pesquisa secundária
  - Busca sistematizar uma área de conhecimento
    - Evolução histórica e estado da arte
  - Metodologias: revisão sistemática ou meta-análise

Taxonomia



Linha do tempo



[1] Primary Research vs Secondary Research - https://research.com/research/primary-research-vs-secondary-research

[2] R. Salles, K. Belloze, F. Porto, P.H. Gonzalez, and E. Ogasawara, 2019, Nonstationary time series transformation methods: An experimental review, *Knowledge-Based Systems*, v. 164, p. 274–291.

Pesquisa quantitativa, qualitativa, analítica e bibliográfica

- Pesquisa Quantitativa
  - Dados numéricos
  - Análises estatísticas

Método

3.1. Protocolo experimental

Visando uma avaliação robusta para a influência das diferentes técnicas de tratamento de *outliers* nas previsões, foi adotada a abordagem de *rolling-origin*. Tal abordagem decompõe a série em janelas móveis de treino e teste, caracterizadas pela atualização sucessiva da origem da previsão e pela produção de previsões para cada nova origem [Hyndman and Athanasopoulos, 2018]. Partindo do conceito de janelas móveis, a cada iteração (atualização da origem da previsão) uma observação é adicionada no final das séries e uma observação é removida no início da série, de modo a permanecer constante o tamanho do conjunto de treinamento. O tamanho da janela de teste é de 15 passos à frente, enquanto a janela de treino contém um ano de observações prévias. As previsões *rolling-origin* fornecem uma melhor compreensão sobre o funcionamento dos modelos pela obtenção de diversas previsões para as séries temporais, viabilizando análise da centralidade e dispersão dos erros de previsões.

Por fim, os efeitos da aplicação dos métodos para o tratamento de *outliers* nas previsões *rolling origin* são avaliados por meio do modelo ARIMA, cuja base reside na hipótese de que as séries temporais são geradas a partir de processo autorregressivo e procuram modelar o processo das séries a fim de prever os seus valores futuros [Kumar and Patil, 2018] [3]. Nos modelos ARIMA o valor corrente de uma série temporal x_t pode ser descrita como uma função de seus p valores passados, $x_{t-1}, \ldots, x_{t-p}$ e de seus q valores de ruído branco passados, $\omega_{t-1}, \ldots, \omega_{t-q}$, de modo que sua forma geral pode ser representada como $\phi(B)(1-B)^d x_t = \theta(B)\omega_t$, onde B é o operador de defasagem, $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$ é o operador autorregressivo (AR) e $\theta(B) = 1 + \theta_1 B + \cdots + \theta_p B^q$ é o operador de médias móveis (MA) [Shumway and Stoffer, 2017].

Como medida de acurácia, foi utilizado o Erro Médio Absoluto Relativo (rMAE), que permite comparações das predições dos modelos candidatos contra predições de modelos de referência. O rMAE pode ser descrito como $\text{rMAE} = (\text{MAE}_a/\text{MAE}_b)$, onde MAE_a é o erro médio absoluto do modelo analisado e MAE_b corresponde ao erro médio absoluto do modelo de referência [Davydenko and Fildes, 2013]. No escopo deste trabalho, o modelo de referência foi gerado a partir das séries temporais sem aplicação de método para tratamento de *outliers*. Assim, quanto mais próximo de zero estiver o rMAE, melhor será o desempenho preditivo do modelo, indicando que o mesmo erra menos em suas previsões do que o próprio modelo de referência. Por outro lado, modelos com rMAE ≥ 1 apresentam piores desempenhos preditivos.

3.2. Resultados

Foram aplicados os diferentes métodos de tratamento de *outliers* ao conjunto de dados. Ao final de cada tratamento, foram analisadas a remanescência de *outliers* nas séries por meio da análise por boxplot. Pela Tabela 1 é possível observar que a regra de alcance interquartil e a winsorização removeram a totalidade dos *outliers* em todas as séries de retorno. A suavização por médias móveis realizou a remoção completa na maioria das séries analisadas (com exceção de BSESN e FTSE).

A Tabela 2 mostra a média e o desvio padrão do rMAE para as previsões realizadas por meio do método ARIMA para as séries de retorno de índices de ações, influenciada

[3] São utilizadas funções que selecionam automaticamente um modelo ARIMA otimizado [Hyndman and Khandakar, 2008].

Resultados

Tabela 1. Desempenho de cada método na remoção de *outliers*

	BVSP	IMOEX	BSESN	IPSA	MXX	HSCE	GSPTSE	FCHI	GDAXI	N225	FTSE	GSPC
Regra de alcance interquartil	X	X	X	X	X	X	X	X	X	X	X	X
Winsorização	X	X	X	X	X	X	X	X	X	X	X	X
Suavização Exponencial Simples												
Suavização Exponencial Quadrática												
Suavização por médias móveis	X	X		X	X	X	X	X	X	X		X
Lowess Smoothing	X											
Filtro HP												
Filtro Recursivo												
Ajustamento Sazonal												

Nota: Cada lacuna preenchida com X indica que houve a remoção completa dos *ouliers*

Tabela 2. Avaliações do rMAE (média e desvio padrão) por tratamento de *outliers*

	BVSP		IMOEX		BSESN		IPSA		MXX		HSCE	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Regra de alcance interquartil	0.944	2.261	0.932	1.420	0.897	1.710	0.926	1.160	0.911	1.529	0.916	2.281
Winsorização	0.871	2.036	0.863	1.284	0.842	1.576	0.858	1.052	0.853	1.406	0.853	2.085
Suavização Exponencial Simples	0.155	0.423	0.174	0.291	0.169	0.340	0.186	0.242	0.168	0.334	0.163	0.423
Suavização Exponencial Quadrática	0.091	0.277	0.090	0.147	0.093	0.189	0.099	0.126	0.097	0.225	0.091	0.250
Suavização por Médias Móveis	0.002	0.005	0.004	0.007	0.002	0.005	0.003	0.003	0.002	0.004	0.002	0.005
Lowess Smoothing	0.008	0.019	0.028	0.042	0.009	0.019	0.015	0.019	0.008	0.015	0.009	0.022
Filtro HP	1.000	2.562	1.001	1.663	1.002	2.075	1.003	1.354	1.002	1.798	1.001	2.673
Filtro Recursivo	1.117	2.833	1.188	1.979	1.184	2.420	1.252	1.690	1.182	2.123	1.170	3.097
Ajustamento Sazonal	0.982	2.500	0.980	1.581	0.994	2.020	0.986	1.315	0.985	1.740	0.989	2.604

	GSPTSE		FCHI		GDAXI		N225		FTSE		GSPC	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Regra de alcance interquartil	0.868	1.216	0.901	1.680	0.903	1.771	0.927	1.827	0.0012	1.687	0.863	1.382
Winsorização	0.820	1.129	0.849	1.558	0.857	1.657	0.868	1.683	0.900	1.388	0.825	1.302
Suavização Exponencial Simples	0.152	0.238	0.149	0.320	0.153	0.328	0.155	0.319	0.850	1.288	0.143	0.268
Suavização Exponencial Quadrática	0.080	0.126	0.082	0.194	0.083	0.183	0.082	0.176	0.150	0.248	0.080	0.165
Suavização por Médias Móveis	0.001	0.002	0.001	0.002	0.001	0.003	0.001	0.003	0.079	0.135	0.001	0.002
Lowess Smoothing	0.009	0.012	0.007	0.016	0.007	0.015	0.009	0.017	0.006	0.011	0.007	0.011
Filtro HP	1.002	1.621	1.001	2.023	1.001	2.112	1.000	2.108	1.001	1.688	1.002	1.822
Filtro Recursivo	1.147	1.809	1.123	2.261	1.134	2.399	1.130	2.373	1.128	1.889	1.095	1.951
Ajustamento Sazonal	1.008	1.587	0.991	1.971	0.990	2.051	0.986	2.049	0.989	1.645	1.001	1.775

pelos diferentes tipos de tratamento de *outliers*. Os resultados mostram que, dentre os métodos que conseguiram remover a totalidade dos *outliers*, a suavização por médias móveis apresentou rMAE próximo de zero e menor dispersão na série de erro de previsão (com exceção de FTSE). Por outro lado, a regra de alcance interquartil apresentou o pior desempenho preditivo.

Com relação aos métodos que não conseguiram remover totalmente os *outliers*, *lowess smoothing* apresentou o melhor desempenho preditivo, com menor rMAE e menor dispersão na série de erro de previsão, em detrimento do filtro recursivo, que apresentou o pior desempenho. Por outro lado, em todos os casos analisados, os métodos de suavização de séries temporais apresentam previsões com desempenho superior às dos métodos de decomposição e filtragem de séries temporais, com rMAE ≥ 1, indicando que tais modelos erraram mais em suas previsões do que os modelos de referência. Diante destes resultados, os *outliers* existentes nas séries financeiras não podem ser considerados erros de medida e que o método utilizado para os tratar interfere na capacidade do modelo de prever a trajetória futura dos retornos de índices de ações.

Análise de métodos de tratamento de outliers para predição dos retornos de índices de ações negociados em bolsa[1]

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Resumo. *Definidos como pontos extremos, os outliers são observações que não seguem um comportamento padrão em uma série temporal. Contudo, esta anomalia pode levar a especificação incorreta do modelo, estimativas viesadas dos parâmetros e previsões de baixa acurácia. Apesar de ser visto como um erro de medida, nas séries financeiras os outliers carregam informações relevantes sobre a dinâmica do mercado acionário e de fatores internacionais. Diante disso, este artigo propõe uma análise comparativa do desempenho de técnicas de tratamento de outliers nas previsões dos índices de ações. Os resultados apontam que métodos com suavização apresentam melhor desempenho, corroborando com a hipótese de que os outliers possuem conteúdo informacional relevante para a previsão da dinâmica dos índices de ações.*

Abstract. *Defined as extreme points, outliers are observations that do not follow a standard behavior in a time series. However, this anomaly can lead to incorrect model specifications, biased parameter estimates, and low accuracy predictions. Despite being seen as a measurement error, in financial series, outliers carry relevant information about the stock market dynamics and international factors. Therefore, this article proposes a comparative analysis of performing outliers treatment techniques in stock index forecasts. The results show that methods with smoothing perform better, corroborating the hypothesis that outliers have relevant informational content for predicting the dynamics of stock prices.*

1. Introdução

Conceituados como pontos extremos existentes em uma série temporal, os *outliers* representam observações que se desviam acentuadamente do comportamento usual das demais observações em um conjunto de dados. Em séries financeiras, os *outliers* podem ser compreendidos como variações anormais nos preços de ações, indicando a existência de fatores que não podem ser modelados comercialmente [Pimento et al., 2018]. Considerado como uma anomalia existente nas séries temporais, os *outliers* podem levar a especificação incorreta de modelos, estimativas viesadas dos parâmetros e avaliação incorreta das previsões, pois interfere na relação entre as observações passadas e futuras. Visto que tais fatores estão associados à natureza altamente ruidosa, volátil e dinâmica das séries temporais financeiras, prever o comportamento futuro do mercado acionário torna-se uma tarefa desafiadora.

[1]Os autores agradecem à FAPERJ, à CAPES (código 001) e ao CNPq pelo financiamento do projeto.

[1] J. Wainer, 2007, "Métodos de pesquisa quantitativa e qualitativa para a Ciência da Computação", *Jornada de Atualização em Informática*, , chapter 1, SBC, p. 32–33.

[2] C. Gea, J. Lima, E. Bezerra, and E. Ogasawara, 2021, Análise de métodos de tratamento de outliers para predição dos retornos de índices de ações negociados em bolsa, In: *Anais do Simpósio Brasileiro de Banco de Dados (SBBD)*, p. 277–282

Pesquisa quantitativa, qualitativa, analítica e bibliográfica

- Pesquisa Qualitativa
  - Dados textuais
  - Ausência de testes estatísticos

Projeto

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studies can be criticized for obtaining small samples sizes, being culturally specific, and/or using pseudo-experimental methods in artificial settings recruiting students. There are many quantitative studies but very few qualitative studies. Further research focusing on the experiences of gamers that allows the gamers to speak and provide their own accounts of playing MMORPGs is needed. The present study explored the attitudes, experiences, and feelings of online gamers through qualitative interviews that made use of gamers' preferred methods of communicating (e-mail and instant messaging).

Method

Participants

Seventy-one online gamers (52 males, 19 females) participated in the study. The participants ranged in age from 18 to 54 years ($M = 26$ years, $SD = 8.4$ years). Most of the participants were from the United States ($n = 32$), followed by the United Kingdom ($n = 19$), Canada ($n = 5$), and Netherlands ($n = 4$), although seven other countries (Norway, Sweden, Portugal, Finland, Australia, Belgium, and Greece) were also represented among the remaining participants ($n = 11$). The mean gaming time per week was 18.9 hours. The mean years of gaming experience was 4.7 years. Gamers were categorized into three gamer types: (a) casual gamer (plays 15 hours a week or less; $n = 39$); (b) regular gamer (plays more than 15 hours and up to 30 hours a week; $n = 21$); (b) excessive gamer (plays more than 30 hours a week; $n = 12$).

Design and procedure

Participants were recruited in response to recruitment posts on various online gaming forums and in-game posts in the World of Warcraft MMORPG. The sample was therefore self-selecting. Participation was voluntary, and no incentive was offered for participation. Semi-structured interviews were conducted over a 4-month period (October 2007 to January 2008). The interviews were carried out (synchronously) via MSN Messenger (an online chat facility) or (asynchronously) via e-mail. All interviews were conducted at a time that was most convenient for the participants. The interviews began with a few structured questions regarding gamer demographics, then asked unstructured questions about gamers' feelings and experiences concerning their day-to-day playing behavior. The unstructured nature of the interviews allowed gamers to develop their own narrative by exploring their experiences of MMORPGs. The researcher allowed gamers to speak for themselves (i.e., the emergent themes were participant led rather than researcher led). This allowed gamers to take control of the interview process and prevented researchers' subjective bias entering the analytic stage. All participants were debriefed before and after the study. Each interview lasted approximately 75 minutes. Those who were interviewed by e-mail generally gave shorter responses to interviewer questions than did those interviewed using MSN Messenger. However, there was no difference in the type of response content. Data were gathered using the preferred medium of the gamer (i.e., e-mail or MSN Messenger). Data from the interviews were imported into QSR Nvivo, a qualitative data analysis software tool, and were analyzed using thematic analysis.

Thematic analysis is a flexible method for identifying, analyzing, and reporting themes within qualitative data.[19] This method can offer a rich description of the data set and is very useful for summarizing large bodies of data. Thematic analysis also allows for the psychological interpretation of data and can generate unanticipated insights. In the first stage of the thematic analysis, the first author read through all the transcripts twice to become familiar with the data and then searched for the main themes to emerge from the responses to each of the questions asked. The responses were reread with particular attention being paid to the themes arising from the first stage of data analysis. The responses were then collated under the emerging theme headings and were given provisional labels and definitions. The responses were again reread to see if they contained any further relevant information to the provisional themes. The themes were then given their final analytical form and definition and were refined further through systematic examination by the first and second authors. The searching, coding, and labeling of themes was done by the use of QSR Nvivo.

Results

Six main themes emerged from the analyses of the interview transcripts: (a) online gaming and integration into day-to-day lives; (b) online gaming, excessive play, and problems; (c) addiction; (d) psychosocial impact of online gaming; (e) online gaming, dissociation, and time loss; and (f) online gaming and the alleviation of negative feelings and mood states.

Online gaming and integration into day-to-day lives

The theme of online gaming and integration into day-to-day lives revealed how gamers fitted their online gaming into their lives. Many gamers said that they played in their free time after work, school, college, or university ($n = 43$; 33 male, 10 female). Others said they integrated their online gaming by playing with friends ($n = 3$, all male), while some said they played during work hours ($n = 5$; 4 male, 1 female). A few gamers ($n = 3$; 1 male, 2 female) played with their spouses or romantic partners. Following are typical quotes by those who played in their free leisure time:

Extract 1: I leave for work at 7 a.m. and work 8 a.m.–5 p.m. I come home, help the kids with homework (I have teenagers), make sure the chores are done, have dinner, and then we log on after that. (P2, female, age 34, USA, regular gamer)

Extract 2: I play after my kids have gone to bed and I have "free" time to do as I wish. My wife chooses to watch TV; I choose to relax by playing video games. (P8, male, age 37, USA, excessive gamer)

These quotes show how gamers played at specific periods in their day-to-day lives. They played before or after work, after dinner, or after completing certain tasks. The findings indicated that gamers appeared to be very organized and managed their time very well. This is contrary to many beliefs that gamers lose control of their lives. Following are typical quotes of gamers who played with their friends:

Extract 3: WoW [World of Warcraft] takes a lot less priority now I'm in my final year at University. A couple of our friends who live nearby also play, so we try to arrange to meet up in

Análise

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game and complete an instance now and again. (P6, male, age 26, UK, casual gamer)

Extract 4: I play with friends I have in real life—both from home and meeting up with them online, and also going to their house and playing in the same room. (P7, male, age 27, USA, regular gamer)

These responses show how gamers played together with friends and how they made game playing a social activity they could enjoy together in real life or online. It also highlights the social nature of MMORPGs. Here are typical quotes of gamers who played while at work:

Extract 5: I work from home, so that doesn't pose a problem. My husband used to play with me, but quit. He went back to FFXI (Final Fantasy XI) and I play WoW (World of Warcraft). (P8, female, age 23, Canada, regular gamer)

Extract 6: I work from home, so if I don't have enough work to do I often spend the rest of the time playing MMORPGs. I also play after my work day is complete. (P9, male, age 36, USA, excessive gamer)

These quotes show how gamers did not have a problem with their jobs and playing MMORPGs. There was no real conflict between work and leisure time because they worked and played in the same location. Integrating online gaming into their lives was very simple for them. Following are some typical quotes from gamers who said they played with their spouses or romantic partners:

Extract 7: When at all possible, I play with my husband. We enjoy yelling insults at each other while our characters are interacting. Hardest part is finding the time to play while the kids are home. (P10, female, age 27, Canada, casual gamer)

Extract 8: I play with my partner, but this is because I actually met my partner through the game. We've since moved in together, but met online, and originally lived thousands of miles apart. (P11, female, age 22, USA, regular gamer)

These quotes show that gamers may play with their partners. For instance, P11 (Extract 8) originally met her partner online within the MMORPG. This shows how the MMORPG can be used as a medium to communicate and meet people.

Online gaming, excessive play, and problems

A majority of the gamers interviewed stated that their game playing is excessive ($n = 27$; 20 male, 7 female) or was excessive ($n = 13$; 11 male, 2 female). Only a few gamers said that their game playing was not excessive ($n = 4$; 2 male, 2 female). The remainder of gamers did not comment on this issue. There was a high degree of awareness of game playing being excessive for those who said that they played excessively. For example, those whose current playing was excessive typically said the following:

Extract 9: Yes, [my playing is] very excessive. The end-game raiding content is what's keeping me hooked as well as the social aspect. (P15, male, age 18, UK, regular gamer)

Extract 10: I still feel I spend too much time playing—but sometimes I can't help myself. . . I just love building my gear and chatting. (P7, male, age 27, USA, regular gamer)

Extract 11: I really don't know why I play it so much. But it's where my friends are at, and it's just something to waste time with. I don't see myself as addicted to WoW [World of Warcraft], even though it seems that way. (P16, male, age 18, UK, excessive gamer)

These quotes show how the social aspects of MMORPGs can have the negative impact of causing excessive game playing. The virtual world was seen as a place to waste time, which led to excessive playing for some gamers. The online community also facilitated excessive play. Following are typical quotes from gamers who said their online gaming was (but no longer is) excessive:

Extract 12: It was quite excessive at one point. I wanted to try all the instances, try all the gear, level up as fast as possible. (P6, male, age 26, UK, casual gamer)

Extract 13: I was "raiding" in one of the best guilds in World of Warcraft. I felt I had the time for it, so I continued even if it required me to play around five to eight hours each day. So I have had a time it was excessive. (P17, male, age 21, Sweden, excessive gamer)

Extract 14: There were a couple of things that contributed to my excessive playing in the past. Foremost among them was belonging to a raiding guild. A second contributing factor was setting goals for myself. (P18, male, age 40, USA, regular gamer)

These quotes provide insight into the nature of MMORPGs and what caused many of the gamers to play excessively in the past. Gamers talked about the raiding, leveling up, and belonging to a guild as factors that caused excessive playing. Also, the quotes provide an insight into the thinking processes of the gamers (e.g., Extract 14, which explain parts of his setting goals for myself. (P18, male, age 40, USA, regular gamer) quest had to be completed). There was a sense of urgency to complete the quest. Following are some quotes from gamers who said that their online gaming was not excessive:

Extract 15: I don't think my MMO playing is excessive. It's a hobby. (P19, female, age 20, USA, regular gamer)

Extract 16: Not really [game playing is not excessive], although I do play a lot. Most from a desire to finish what I started. (P21, male, age 21, UK, regular gamer)

Gamers here explained their online gaming behavior as not excessive and viewed it as a hobby. However, P21 (Extract 16) showed signs of playing excessively even though he said his game playing was not excessive.

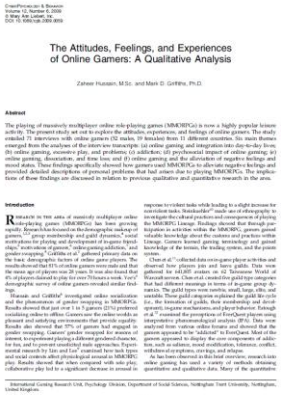
Addiction

Online gaming was seen as potentially addictive by some gamers ($n = 14$; 10 male, 4 female). For instance:

Extract 17: It's obvious that MMORPGs are addictive. My competitive nature is constantly hooked by testing my skill with other people. (P22, female, age 20, USA, casual gamer)

Extract 18: MMORPGs will be the new cocaine and the new trigger of the status of the mind's emotional state. MMORPGs are highly addictive. (P24, female, age 18, UK, regular gamer)

Extract 19: EverQuest used to be called "Evercrack" due to its highly addictive nature and large-scale social interactions. This led people to skip work to play, leading to a loss of a job (I watched it happen at my place of employment). (P23, male, age 27, USA, regular gamer)



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The Attitudes, Feelings, and Experiences of Online Gamers: A Qualitative Analysis

Zaheer Hussain, M.Sc. and Mark D. Griffiths, Ph.D.

Abstract

The playing of massively multiplayer online role-playing games (MMORPGs) is now a highly popular leisure activity. The present study set out to explore the attitudes, experiences, and feelings of online gamers. The study entailed 71 interviews with online gamers (52 males, 19 females) from 11 different countries. Six main themes emerged from the analyses of the interview transcripts: (a) online gaming and integration into day-to-day lives; (b) online gaming, excessive play, and problems; (c) addiction; (d) psychosocial impact of online gaming; (e) online gaming, dissociation, and time loss; and (f) online gaming and the alleviation of negative feelings and mood states. These findings specifically showed how gamers used MMORPGs to alleviate negative feelings and provided detailed descriptions of personal problems that had arisen due to playing MMORPGs. The implications of these findings are discussed in relation to previous qualitative and quantitative research in the area.

Introduction

Research in the area of massively multiplayer online role-playing games (MMORPGs) has been growing rapidly. Research has focused on the demographic makeup of gamers,[1–7] group membership and guild dynamics,[8] social capital,[9] the impact of online gaming on offline relationships and friendships,[10] addiction,[11] problem gaming addiction,[12] and gender swapping.[13] Griffiths et al.[14] gathered primary data on the basic demographic factors of online game players. The results showed that 81% of online gamers were male and that the mean age of players was 28 years. It was also found that 4% of players claimed to play for over 70 hours a week. Yee's[15] demographic survey of online gamers revealed similar findings.

Hussain and Griffiths[16] investigated online socialization and the phenomenon of gender swapping in MMORPGs. Results showed that just over 1 in 5 gamers (21%) preferred socializing online to offline. Gamers saw the online world as a pleasant and satisfying environments that provide equality. Results also showed that 77% of gamers had engaged in gender swapping. Gamers' gender swapped for reasons of anonymity, to enhance their game playing experiences, for fun, and to prevent unsolicited male approaches. Experimental research by Yee and Lee[17] examined how task types and social contexts affect physiological arousal in MMORPG play. Results showed that when compared with solo play, collaborative play led to a significant decrease in arousal in response to violent tasks while leading to a slight increase for non-violent tasks. Steinkuehler[18] made use of ethnography to investigate the cultural practices and consequences of playing the MMORPG Lineage. Findings showed that through participation in activities within the MMORPG, gamers gained valuable knowledge about the customs and practices within the game. Gamers learned gaming terminology and gained knowledge of the terrain, the leveling system, and the points system.

Chen et al.[19] collected data on a gamer player activities and observed how players join and leave guilds. Data were gathered for 140,000 avatars on 12 Taiwanese World of Warcraft servers. Chen et al. created five guild type categories that had different meanings in terms of in-game group dynamics. The guild types were neutrals, small, large, elite, and mixed. Results showed that gamers' behavior was dependent on (i.e., the formation of guilds, their membership and development), in-game mechanisms, and player behavior. Fabricatore et al.[20] examined the perceptions of EverQuest players using interpretative phenomenological analysis (IPA). Data were analyzed from various online forums and showed that the gamers appeared to be "addicted" to EverQuest. Most of the gamers experienced a "rush" from the game and had positive feelings, such as calmness, mood modification, tolerance, conflict, withdrawal symptoms, cravings, and relapses.

As has been observed in this brief overview, research into online gaming has used a variety of methods obtaining quantitative and qualitative data. Many of the quantitative

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[1] J. Wainer, 2007, "Métodos de pesquisa quantitativa e qualitativa para a Ciência da Computação", *Jornada de Atualização em Informática*, , chapter 1, SBC, p. 32–33.

[2] Z. Hussain and M.D. Griffiths, 2009, The Attitudes, Feelings, and Experiences of Online Gamers: A Qualitative Analysis, CyberPsychology & Behavior, v. 12, n. 6 (Dezembro.), p. 747–753.

Pesquisa quantitativa, qualitativa e analítica

- Pesquisa Analítica
  - Construção de teorias e apresentação de provas

Algoritmo de coloração de grafos

An upper bound for the chromatic number of a graph and its application to timetabling problems

By D. J. A. Welsh and M. B. Powell*

This paper points out the connection between the basic scheduling or timetabling problem with the well known problem of colouring the vertices of a graph in such a way that (i) no two adjacent vertices are the same colour and (ii) the number of colours used is a minimum. We give an algorithm for colouring a graph subject to (i) and give a new easily determined bound for the number of colours needed. This same bound is also a new upper bound for the chromatic number of a graph in terms of the degrees of its vertices.

1. Introduction

The basic scheduling problem may be described as follows. We are given a jobs $\{J_i\}_{i=1}^{n}$, together with an incompatibility matrix $M \equiv \{m_{ij}\}$, where m_{ij} = zero or 1 depending on whether or not J_i and J_j can be carried out on the same day. The problem is to find the minimum number of days needed to carry out the n jobs.

By representing each job J_i by a point A_i and joining A_i to A_j by an undirected edge e_{ij} if and only if $m_{ij} = 1$, the above problem is clearly seen to be equivalent to finding a minimum colouring of the vertices of the graph G which has vertex set $V(G) \equiv \{A_i\}_{i=1}^{n}$ and edge set $E(G) \equiv \{e_{ij}\}$.

In the notation of Ore (1962), the minimum number of days needed is exactly the chromatic number $k(G)$, of the graph G. The problem of determining this chromatic number for an arbitrary graph G is a well known unsolved problem. In theory a solution can always be obtained by trial and error or by integer programming methods. However, in practical instances such as arise in large scale scheduling problems the labour involved often renders the method of solution inefficient, and as a compromise a solution is sought which blends the advantages of simplicity and reasonable accuracy.

2. Statement of results

The degree of a vertex A_i of the graph G is the number of edges having A_i as an endpoint, and we will denote it by d_i. Without loss of generality we assume that

$$d_1 \geqslant d_2 \geqslant \ldots \geqslant d_n. \tag{1}$$

It is easy to show (Ore 1962, Chapter 14) that if $k(G)$ denotes the chromatic number of G then

$$k(G) \leqslant d_1 + 1 \tag{2}$$

and provided G contains no d_1-simplices, then from (2) we know

$$k(G) \leqslant d_1. \tag{3}$$

The purpose of this paper is to point out that G may always be coloured in at most $\alpha(G)$ colours where

$$k(G) \leqslant \alpha(G) \leqslant \max_i \min \begin{bmatrix} d_i + 1 \\ i \end{bmatrix}. \tag{4}$$

Furthermore, for those instances where this number is deemed sufficiently economical, a colouring employing no more than this number of colours may be obtained by a very simple algorithm described below.

The algorithm. Let $V(G)$ denote the vertex set of G. Let T_1 be a subset of $V(G)$ defined recursively as follows:

$$A_i \in T_1.$$

If $\{A_{i_k}\}_{k=1}^{m} \in T_1$, where $i_1 = 1$ and

$$i_1 < i_2 < \ldots < i_m$$

then $A_j \in T_1$ if and only if

$$j > i_m$$

and also A_j is not adjacent to any member of $\{A_{i_k}\}_{k=1}^{m}$.

Clearly T_1 is a finite, non-null subset of $V(G)$. Similarly we define T_2 to be a subset of $V(G) - T_1$ constructed recursively by:

If i is the least integer for which $A_i \notin T_1$, then $A_i \in T_2$. If $\{A_{j_k}\}_{k=1}^{p} \in T_2$ where $j_1 < j_2 < \ldots < j_p$ then $A_1 \in T_2$ if and only if

$$1 > j_p$$

and A_1 is not linked in G to any of the vertices $\{A_{j_k}\}_{k=1}^{p}$. Notice that T_2 may be null (when G is completely disconnected). Similarly we define T_3 to be a subset of $V(G) - (T_1 \cup T_2)$ such that: If i is the least integer for which $A_i \notin T_1 \cup T_2$, then $A_i \in T_3$, and then continuing as before.

In this way we define a sequence $\{T_i\}$ of disjoint subsets of $V(G)$ such that $V(G) = \bigcup_{i=1}^{m} T_i$ and there exists a finite integer $\alpha(G)$ such that

$$T_r = \phi \quad r > \alpha(G).$$

From the nature of the construction no two vertices in a given set T_r are adjacent in G and hence by assigning

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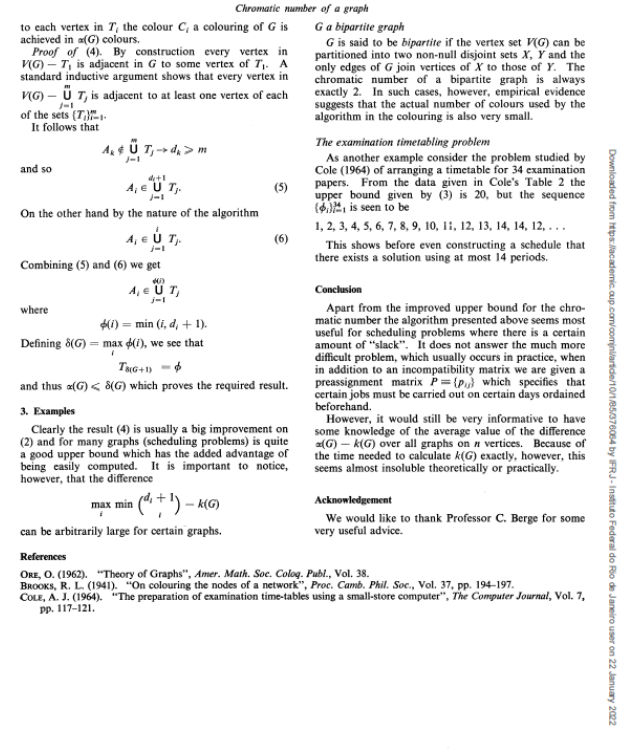

to each vertex in T_i the colour C_i a colouring of G is achieved in $\alpha(G)$ colours.

Proof of (4). By construction every vertex in $V(G) - T_1$ is adjacent in G to some vertex of T_1. A standard inductive argument shows that every vertex in $V(G) - \bigcup_{j=1}^{m} T_j$ is adjacent to at least one vertex of each of the sets $\{T_j\}_{j=1}^{m}$.

It follows that

$$A_k \notin \bigcup_{j=1}^{m} T_j \rightarrow d_k \geqslant m$$

and so

$$A_i \in \bigcup_{j=1}^{d_i+1} T_j. \tag{5}$$

On the other hand by the nature of the algorithm

$$A_i \in \bigcup_{j=1}^{i} T_j. \tag{6}$$

Combining (5) and (6) we get

$$A_i \in \bigcup_{j=1}^{\phi(i)} T_j$$

where

$$\phi(i) = \min(i, d_i + 1).$$

Defining $\delta(G) = \max_i \phi(i)$, we see that

$$T_{\delta(G+1)} = \phi$$

and thus $\alpha(G) \leqslant \delta(G)$ which proves the required result.

3. Examples

Clearly the result (4) is usually a big improvement on (2) and for many graphs (scheduling problems) is quite a good upper bound which has the added advantage of being easily computed. It is important to notice, however, that the difference

$$\max_i \min \binom{d_i + 1}{i} - k(G)$$

can be arbitrarily large for certain graphs.

G a bipartite graph

G is said to be *bipartite* if the vertex set $V(G)$ can be partitioned into two non-null disjoint sets X, Y and the only edges of G join vertices of X to those of Y. The chromatic number of a bipartite graph is always exactly 2. In such cases, however, empirical evidence suggests that the actual number of colours used by the algorithm in the colouring is also very small.

The examination timetabling problem

As another example consider the problem studied by Cole (1964) of arranging a timetable for 34 examination papers. From the data given in Cole's Table 2 the upper bound given by (3) is 20, but the sequence $\{\phi_i\}_{i=1}^{34}$ is seen to be

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 14, 12, . . .

This shows before even constructing a schedule that there exists a solution using at most 14 periods.

Conclusion

Apart from the improved upper bound for the chromatic number the algorithm presented above seems most useful for scheduling problems where there is a certain amount of "slack". It does not answer the much more difficult problem, which usually occurs in practice, when in addition to an incompatibility matrix we are given a preassignment matrix $P \equiv \{p_{ij}\}$ which specifies that certain jobs must be carried out on certain days ordained beforehand.

However, it would still be very informative to have some knowledge of the average value of the difference $\alpha(G) - k(G)$ over all graphs on n vertices. Because of the time needed to calculate $k(G)$ exactly, however, this seems almost insoluble theoretically or practically.

Acknowledgement

We would like to thank Professor C. Berge for some very useful advice.

References

Ore, O. (1962). "Theory of Graphs", *Amer. Math. Soc. Coloq. Publ.*, Vol. 38.
Brooks, R. L. (1941). "On colouring the nodes of a network", *Proc. Camb. Phil. Soc.*, Vol. 37, pp. 194–197.
Cole, A. J. (1964). "The preparation of examination time-tables using a small-store computer", *The Computer Journal*, Vol. 7, pp. 117–121.

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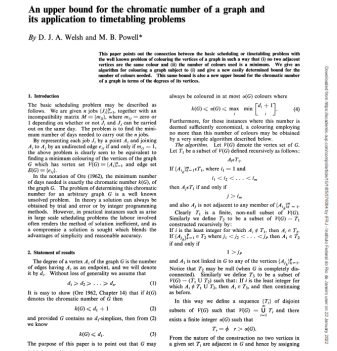

An upper bound for the chromatic number of a graph and its application to timetabling problems

By D. J. A. Welsh and M. B. Powell*

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1. Introduction

2. Statement of results

Conclusion

Acknowledgement

References

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[1] J. Wainer, 2007, "Métodos de pesquisa quantitativa e qualitativa para a Ciência da Computação", *Jornada de Atualização em Informática*, , chapter 1, SBC, p. 32–33.
[2] D.J.A. Welsh and M.B. Powell, 1967, An upper bound for the chromatic number of a graph and its application to timetabling problems, The Computer Journal, v. 10, n. 1 (Jan.), p. 85–86.

Apresentação de algo diferente

- Análise qualitativa
  - Apresentar que não há trabalhos relacionados
  - Realizar avaliações qualitativas
    - Não for possível realizar avaliações quantitativas
  - Estudos de caso usualmente não prova
    - Típico de áreas onde é difícil conseguir dados
    - Uma boa hipótese e argumentação para ser convincente

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SPECIAL ISSUE PAPER

ProvManager: a provenance management system for scientific workflows

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SUMMARY

Running scientific workflows in distributed and heterogeneous environments has been a motivating approach for provenance management, which is loosely coupled to the workflow execution engine. This kind of approach is interesting because it allows both storage and access to provenance data in a heterogeneous environment and does not impose any overhead on workflow execution performance. However, most of the existing approaches overload scientists with many ad-hoc tasks, such as script adaptations and implementations of extra functionalities to provide provenance independence. This paper proposes ProvManager, a provenance management approach that eases the gathering, storage, and analysis of provenance information in a distributed and heterogeneous environment scenario, without putting the burden of adaptations on the scientist. ProvManager leverages multiple workflow management systems (Vistrails and Kepler) and relies on workflow iterations from multiple workflow management systems. Copyright © 2011 John Wiley & Sons, Ltd.

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KEY WORDS: provenance management; scientific experiment; scientific workflow

1. INTRODUCTION

High-performance computing and scientific workflows have evolved as important resources to scientists, assisting in simulation and interpretation of experimental data. As a result, they provide important leads for further experimental research.

The workflow concept is used to support scientific experiments based on simulations, allowing a structured composition of programs as a sequence of activities that aims at a specific result [1]. Afterwards, scientific workflow management systems (SWfMS) were envisioned to control the orchestration and execution of these so-called scientific workflows. SWfMS not only make the steps of the workflow execution explicit, but also enable an automatic collection of workflow data products and their provenance information [2].

Provenance data provide historical information about data manipulated in a workflow [3]. This historical information expresses how data products were generated, showing their transformation processes from primary input and intermediary data. The management of this kind of information is extremely important because it provides scientists with a variety of data analysis applications. For instance, from provenance information it is possible to verify data quality of generated data products

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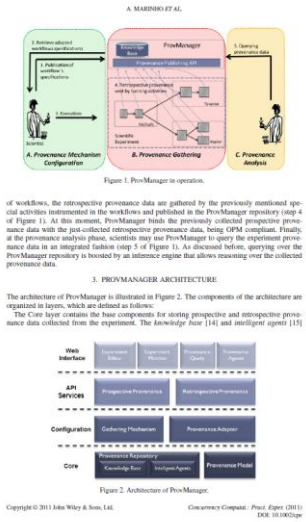
Contexto



A. MARINHO *ET AL.*

Figure 1. ProvManager in operation.

of workflows, the retrospective provenance data are gathered by the previously mentioned special activities instrumented in the workflows and published in the ProvManager repository (step 4 of Figure 1). At this moment, ProvManager binds the previously collected prospective provenance data with the just-collected retrospective provenance data, being OPM compliant. Finally, at the provenance analysis phase, scientists may use ProvManager to query the experiment provenance data in an integrated fashion (step 5 of Figure 1). As discussed before, querying over the ProvManager repository is boosted by an inference engine that allows reasoning over the collected provenance data.

3. PROVMANAGER ARCHITECTURE

The architecture of ProvManager is illustrated in Figure 2. The components of the architecture are organized in layers, which are defined as follows:

The Core layer contains the base components for storing prospective and retrospective provenance data collected from the experiment. The *knowledge base* [14] and *intelligent agents* [15]

Figure 2. Architecture of ProvManager.

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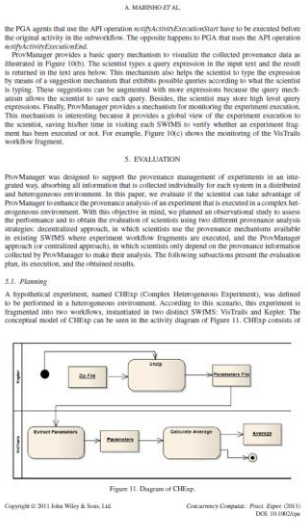
Argumentação



A. MARINHO *ET AL.*

the PGA agents that use the API operation *notifyActivityExecutionStart* have to be executed before the original activity in the subworkflow. The opposite happens to PGA that uses the API operation *notifyActivityExecutionEnd*.

ProvManager provides a basic query mechanism to visualize the collected provenance data as illustrated in Figure 10(b). The scientist types a query expression in the input text and the result is returned in the text area below. This mechanism also helps the scientist to type the expression by means of a suggestion mechanism that exhibits possible queries according to what the scientist is typing. These suggestions can be augmented with more expressions because the query mechanism allows the scientist to save each query. Besides, the scientist may store high level query expressions. Finally, ProvManager provides a mechanism for monitoring the experiment execution. This mechanism is interesting because it provides a global view of the experiment execution to the scientist, saving his/her time in visiting each SWfMS to verify whether an experiment fragment has been executed or not. For example, Figure 10(c) shows the monitoring of the VisTrails workflow fragment.

5. EVALUATION

ProvManager was designed to support the provenance management of experiments in an integrated way, absorbing all information that is collected individually for each system in a distributed and heterogeneous environment. In this paper, we evaluate if the scientist can take advantage of ProvManager to enhance the provenance analysis of an experiment that is executed in a complex heterogeneous environment. With this objective in mind, we planned an observational study to assess the performance and to obtain the evaluation of scientists using two different provenance analysis strategies: decentralized approach, in which scientists use the provenance mechanisms available in existing SWfMS where experiment workflow fragments are executed, and the ProvManager approach (or centralized approach), in which scientists only depend on the provenance information collected by ProvManager to make their analysis. The following subsections present the evaluation plan, its execution, and the obtained results.

5.1. Planning

A hypothetical experiment, named CHExp (Complex Heterogeneous Experiment), was defined to be performed in a heterogeneous environment. According to this scenario, this experiment is fragmented into two workflows, instantiated in two distinct SWfMS: VisTrails and Kepler. The conceptual model of CHExp can be seen in the activity diagram of Figure 11. CHExp consists of

Figure 11. Diagram of CHExp.

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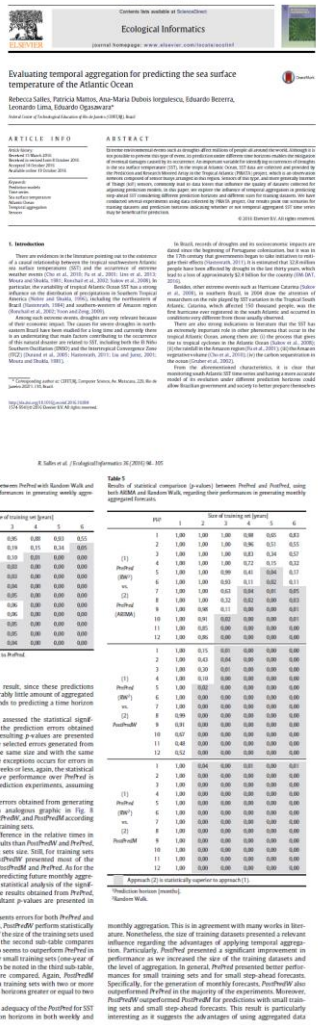
[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.
[2] A. Marinho, L. Murta, C. Werner, V. Braganholo, S.M.S.D. Cruz, E. Ogasawara, and M. Mattoso, 2012, ProvManager: A provenance management system for scientific workflows, *Concurrency and Computation: Practice and Experience*, v. 24, n. 13, p. 1513–1530.

Apresentação de algo melhor

- Exige comparação com a literatura
- Uso de benchmark
  - Analisado por meio de testes padronizados reconhecidos internacionalmente
  - O trabalho se concentra na elaboração da hipótese e não na busca dos dados
- Quando não tem benchmark
  - O próprio autor cria seus testes
  - Trabalho extra e possibilidade de introdução de erros
  - Recodificar o código das abordagens publicadas
- Importante ter uma métrica clara
  - Não é necessário ser melhor em tudo



Ecological Informatics

Evaluating temporal aggregation for predicting the sea surface temperature of the Atlantic Ocean

Rebecca Salles, Patricia Mattos, Ana-Maria Dubois Iorgulescu, Eduardo Bezerra, Leonardo Lima, Eduardo Ogasawara*

ARTICLE INFO

ABSTRACT

1. Introduction

Table 4

Table 5

[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

[2] R. Salles, P. Mattos, A.-M.D. Iorgulescu, E. Bezerra, L. Lima, and E. Ogasawara, 2016, Evaluating temporal aggregation for predicting the sea surface temperature of the Atlantic Ocean, *Ecological Informatics*, v. 36, p. 94–105.

Apresentação de uma prova

- Deve-se construir uma teoria (conjunto de definições) e uma prova formal de seus principais teoremas
- Típico das subáreas ligadas à Lógica e Matemática
- Atualmente exigem experimentação

ELSEVIER

Theoretical Computer Science 188 (1997) 1–44

Theoretical Computer Science

Tutorial

Optimal binary search trees

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Received October 1996

Communicated by M. Nivat

Abstract

We consider the problem of building *optimal binary search trees*. The binary search tree is a widely used data structure for information storage and retrieval. A binary search tree T for a set of keys from a total order is a binary tree in which each node has a key value and all the keys of the left subtree are less than the key at the root and all the keys of the right subtree are greater than the key at the root, this property holding recursively for the left and right subtrees of the tree T.

Suppose we are given n keys and the probabilities of accessing each key and those occurring in the gap between two successive keys. The *optimal binary search tree problem* is to construct a binary search tree on these n keys that minimizes the expected access time. One variant of this problem is when only the gaps have nonzero access probabilities, and is called the *optimal alphabetic tree problem*. Another related problem is when these n re-order between the keys and there are probabilities associated only with the gaps and the objective is to build a binary tree with minimum expected weighted path length from the root. This is called the *Huffman tree problem*.

In this survey, we assess known results on the structural properties of the optimal trees, algorithms and lower bounds to construct and to verify optimal trees and heuristics to construct nearly optimal trees and other related results.

Keywords: Binary search tree; Data structures; Optimal binary search tree

1. Introduction

1.1. Motivation

The *binary search tree* is a widely used data-structure for information storage and retrieval as it supports many dynamic-set operations including Search, Minimum, Maximum, Predecessor, Successor, Insert and Delete.

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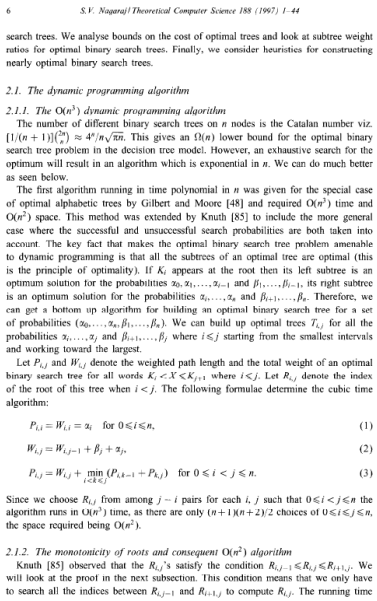
PII S0304-3975(96)00220-X

Formalização



search trees. We analyse bounds on the cost of optimal trees and look at subtree weight ratios for optimal binary search trees. Finally, we consider heuristics for constructing nearly optimal binary search trees.

2.1. The dynamic programming algorithm

2.1.1. The $O(n^3)$ *dynamic programming algorithm*

The number of different binary search trees on n nodes is the Catalan number viz. $[1/(n+1)]\binom{2n}{n} \approx 4^n/n\sqrt{\pi n}$. This gives an $\Omega(n)$ lower bound for the optimal binary search tree problem in the decision tree model. However, an exhaustive search for the optimum will result in an algorithm which is exponential in n. We can do much better as seen below.

The first algorithm running in time polynomial in n was given for the special case of optimal alphabetic trees by Gilbert and Moore [48] and required $O(n^3)$ time and $O(n^2)$ space. This method was extended by Knuth [85] to include the more general case where the successful and unsuccessful search probabilities are both taken into account. The key fact that makes the optimal binary search tree problem amenable to dynamic programming is that all the subtrees of an optimal tree are optimal (this is the principle of optimality). If K_r appears at the root then its left subtree is an optimum solution for the probabilities $\alpha_0, \alpha_1, \ldots, \alpha_{r-1}$ and $\beta_1, \ldots, \beta_{r-1}$, its right subtree is an optimum solution for the probabilities $\alpha_r, \ldots, \alpha_n$ and $\beta_{r+1}, \ldots, \beta_n$. Therefore, we can get a bottom up algorithm for building an optimal binary search tree for a set of probabilities $(\alpha_0, \ldots, \alpha_n, \beta_1, \ldots, \beta_n)$. We can build up optimal trees $T_{i,j}$ for all the probabilities $\alpha_i, \ldots, \alpha_j$ and $\beta_{i+1}, \ldots, \beta_j$ where $i \leqslant j$ starting from the smallest intervals and working toward the largest.

Let $P_{i,j}$ and $W_{i,j}$ denote the weighted path length and the total weight of an optimal binary search tree for all words $K_i <\cdot \mathcal{X} <\cdot K_{j+1}$ where $i \leqslant j$. Let $R_{i,j}$ denote the index of the root of this tree when $i < j$. The following formulae determine the cubic time algorithm:

$$P_{i,i} = W_{i,i} = \alpha_i \quad \text{for } 0 \leqslant i \leqslant n, \tag{1}$$

$$W_{i,j} = W_{i,j-1} + \beta_j + \alpha_j, \tag{2}$$

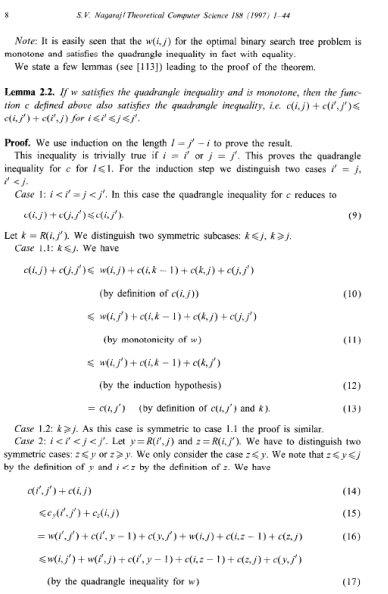
$$P_{i,j} = W_{i,j} + \min_{i<k\leqslant j} (P_{i,k-1} + P_{k,j}) \quad \text{for } 0 \leqslant i < j \leqslant n. \tag{3}$$

Since we choose $R_{i,j}$ from among $j-i$ pairs for each i, j such that $0 \leqslant i < j \leqslant n$ the algorithm runs in $O(n^3)$ time, as there are only $(n+1)(n+2)/2$ choices of $0 \leqslant i \leqslant j \leqslant n$, the space required being $O(n^2)$.

2.1.2. The monotonicity of roots and consequent $O(n^2)$ *algorithm*

Knuth [85] observed that the $R_{i,j}$'s satisfy the condition $R_{i,j-1} \leqslant R_{i,j} \leqslant R_{i+1,j}$. We will look at the proof in the next subsection. This condition means that we only have to search all the indices between $R_{i,j-1}$ and $R_{i+1,j}$ to compute $R_{i,j}$. The running time

Prova



Note: It is easily seen that the $w(i,j)$ for the optimal binary search tree problem is monotone and satisfies the quadrangle inequality in fact with equality.

We state a few lemmas (see [113]) leading to the proof of the theorem.

Lemma 2.2. *If w satisfies the quadrangle inequality and is monotone, then the function c defined above also satisfies the quadrangle inequality, i.e.* $c(i,j) + c(i',j') \leqslant c(i,j') + c(i',j)$ *for* $i \leqslant i' \leqslant j \leqslant j'$.

Proof. We use induction on the length $l = j' - i$ to prove the result.

This inequality is trivially true if $i = i'$ or $j = j'$. This proves the quadrangle inequality for c for $l \leqslant 1$. For the induction step we distinguish two cases $i' = j$, $i' < j$.

Case 1: $i < i' = j < j'$. In this case the quadrangle inequality for c reduces to

$$c(i,j) + c(j,j') \leqslant c(i,j'). \tag{9}$$

Let $k = R(i,i')$. We distinguish two symmetric subcases: $k \leqslant j$, $k \geqslant j$.

Case 1.1: $k \leqslant j$. We have

$$\begin{aligned} c(i,j) + c(j,j') &\leqslant w(i,j) + c(i,k-1) + c(k,j) + c(j,j') \\ &\quad \text{(by definition of } c(i,j)) && (10) \\ &\leqslant w(i,j') + c(i,k-1) + c(k,j) + c(j,j') \\ &\quad \text{(by monotonicity of } w) && (11) \\ &\leqslant w(i,j') + c(i,k-1) + c(k,j') \\ &\quad \text{(by the induction hypothesis)} && (12) \\ &= c(i,j') \quad \text{(by definition of } c(i,j') \text{ and } k). && (13) \end{aligned}$$

Case 1.2: $k \geqslant j$. As this case is symmetric to case 1.1 the proof is similar.

Case 2: $i < i' < j < j'$. Let $y = R(i',j)$ and $z = R(i,j')$. We have to distinguish two symmetric cases: $z \leqslant y$ or $z \geqslant y$. We only consider the case $z \leqslant y$. We note that $z \leqslant y \leqslant j$ by the definition of y and $i < z$ by the definition of z. We have

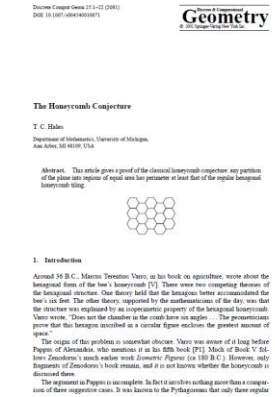
$$\begin{aligned} &c(i',j') + c(i,j) && (14) \\ &\leqslant c_2(i',j') + c_2(i,j) && (15) \\ &= w(i',j') + c(i',y-1) + c(y,j') + w(i,j) + c(i,z-1) + c(z,j) && (16) \\ &\leqslant w(i,j') + w(i',j) + c(i',y-1) + c(i,z-1) + c(z,j) + c(y,j') \\ &\quad \text{(by the quadrangle inequality for } w) && (17) \end{aligned}$$

[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.

[2] S.V. Nagaraj, 1997, Optimal binary search trees, *Theoretical Computer Science*, v. 188, n. 1 (Nov.), p. 1–44.

Ciência e tecnologia

- Ciência busca o conhecimento e as explicações
  - Ciência constrói teorias para explicar fatos observados
- Tecnologia é a aplicação dos conhecimentos nas atividades práticas (industriais e econômicas)
  - Não objetiva explicar o mundo
  - Transforma o mundo
- Ciência e tecnologia caminham juntas na computação
- Trabalhos científicos (mix)
  - Explica o porquê das coisas funcionarem como funcionam
  - Responde uma hipótese de pesquisa
  - Apresenta aplicações práticas (ferramenta, algoritmo, protótipo, processo)
  - Teses, Dissertações e TCCs
    - Apresenta ideias
    - Contribuição para o conhecimento
    - Disponibiliza artefatos


Discrete Comput Geom 25:1–22 (2001)
DOI: 10.1007/s004540010071

Discrete & Computational Geometry

The Honeycomb Conjecture

T. C. Hales

Department of Mathematics, University of Michigan, Ann Arbor, MI 48109, USA

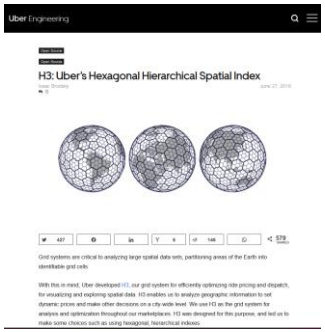
Abstract. This article gives a proof of the classical honeycomb conjecture: any partition of the plane into regions of equal area has perimeter at least that of the regular hexagonal honeycomb tiling.

1. Introduction

Around 36 B.C., Marcus Terentius Varro, in his book on agriculture, wrote about the hexagonal form of the bee's honeycomb [V]. There were two competing theories of the hexagonal structure. One theory held that the hexagons better accommodated the bee's six feet. The other theory, supported by the mathematicians of the day, was that the structure was explained by an isoperimetric property of the hexagonal honeycomb. Varro wrote, "Does not the chamber in the comb have six angles . . . The geometricians prove that this hexagon inscribed in a circular figure encloses the greatest amount of space."

The origin of this problem is somewhat obscure. Varro was aware of it long before Pappus of Alexandria, who mentions it as his fifth book [P1]. Much of Book V follows Zenodorus's much earlier work *Isometric Figures* (ca 180 B.C.). However, only fragments of Zenodorus's book remain, and it is not known whether the honeycomb is discussed there.

The argument in Pappus is incomplete. In fact it involves nothing more than a comparison of three suggestive cases. It was known to the Pythagoreans that only three regular


Uber Engineering

H3: Uber's Hexagonal Hierarchical Spatial Index

Grid systems are critical to analyzing large spatial data sets, partitioning areas of the Earth into identifiable grid cells.

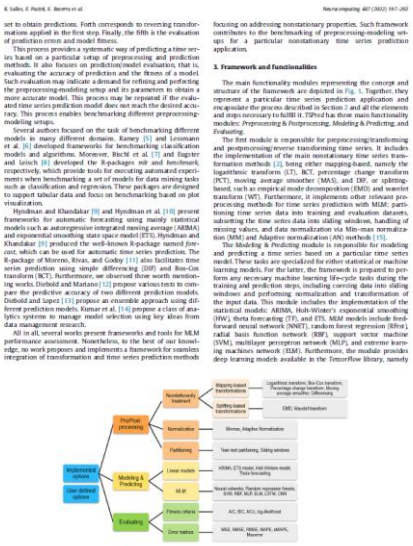
With this in mind, Uber developed H3, our grid system for efficiently optimizing ride pricing and dispatch, for visualizing and exploring spatial data. H3 enables us to analyze geographic information to set dynamic prices and make other decisions on a city-wide level. We use H3 as the grid system for analysis and optimization throughout our marketplaces. H3 was designed for this purpose, and led us to make some choices such as using hexagonal, hierarchical indexes.

[1] T.C. Hales, 2001, The Honeycomb Conjecture, *Discrete & Computational Geometry*, v. 25, n. 1 (Jan.), p. 1–22.
[2] I. Brodsky, 2018. H3: Uber's Hexagonal Hierarchical Spatial Index. *Uber Engineering Blog*. URL: https://eng.uber.com/h3

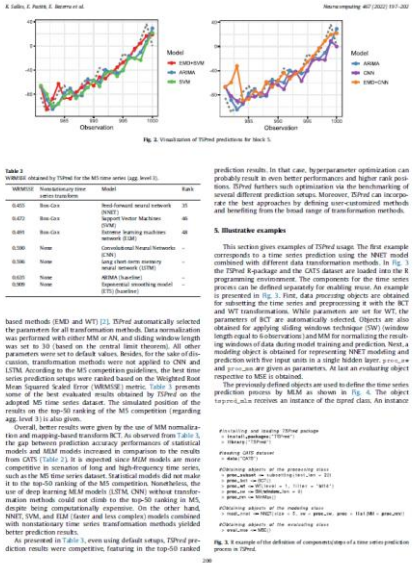
Apresentação de um artefato computacional

- "Fiz algo novo: Eis meu produto"
  - Incomum em áreas maduras
  - Nem sempre há rigor científico na apresentação dos resultados
  - Apropriado para seção de ferramentas
  - Bom escopo para TCC

Arquitetura



Uso



[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.
[2] R. Salles, E. Pacitti, E. Bezerra, F. Porto, and E. Ogasawara, 2021, TSPred: A framework for nonstationary time series prediction, *Neurocomputing*, v. 467, p. 197–202.

Apresentação de artigos de dados

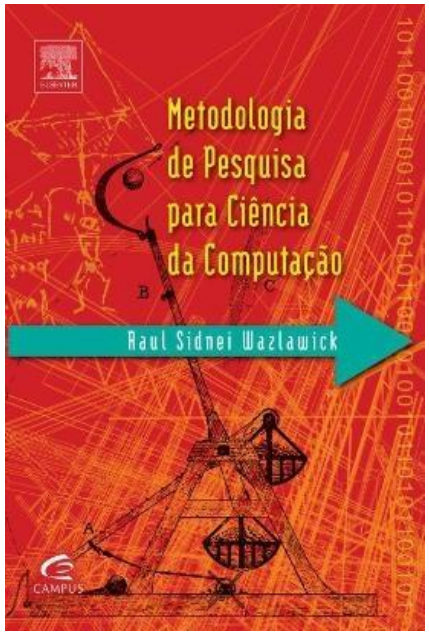
- Um artigo de dados é um documento revisado por pares que descreve um conjunto de dados
- Valoriza o esforço para preparar, organizar e descrever dados

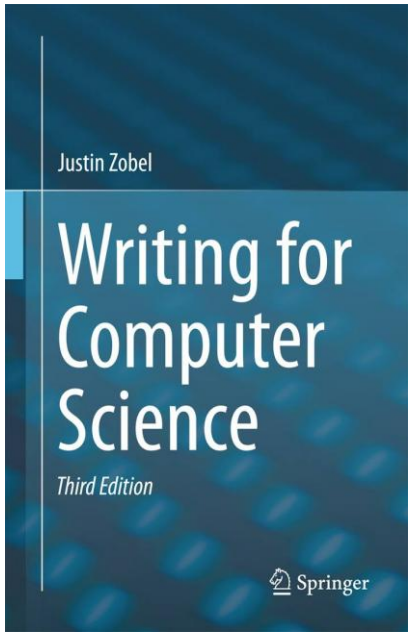
ETL

Uso

[1] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.
[2] R. Salles, E. Pacitti, E. Bezerra, F. Porto, and E. Ogasawara, 2021, TSPred: A framework for nonstationary time series prediction, *Neurocomputing*, v. 467, p. 197–202.

Referências





Leitura adicional: Metodologia de Pesquisa para Ciência da Computação, Capítulos 3 a 5

Slides e vídeos em: https://eic.cefet-rj.br/~eogasawara/metodologia-cientifica/

[1] J. Zobel, 2015, *Writing for Computer Science*. Springer.
[2] R. Wazlawick, 2017, *Metodologia de Pesquisa para Ciência da Computação*. Elsevier Brasil.